Image Classification Report

Random Forest vs. SVM on a 5-Class Image Dataset

Dataset	Preprocessing
• images.zip • 5 classes • total: 309 • split: 247/62	• grayscale • resize to (16, 16) • normalize to [0,1] • flatten pixels

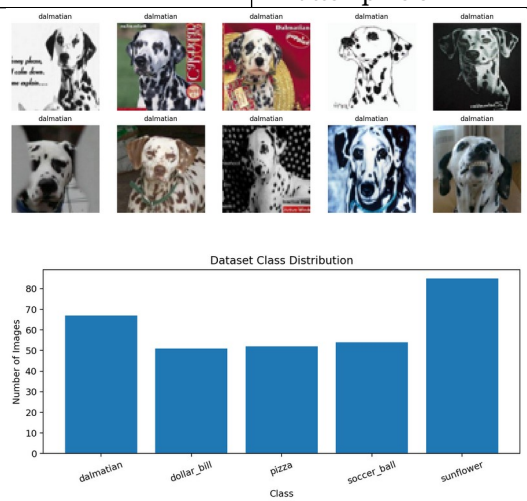

Figure 1. Sample images and class distribution.

Methodology

Random Forest used GridSearchCV across n_estimators, max_depth, min_samples_split, and min_samples_leaf. Best settings: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 5, 'n_estimators': 100}. SVM used linear and RBF kernels with different C values. Best settings: {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}.

Results

Model	Acc.	Prec.	Recall	F1
Random Forest	0.710	0.744	0.710	0.705
SVM	0.645	0.669	0.645	0.648

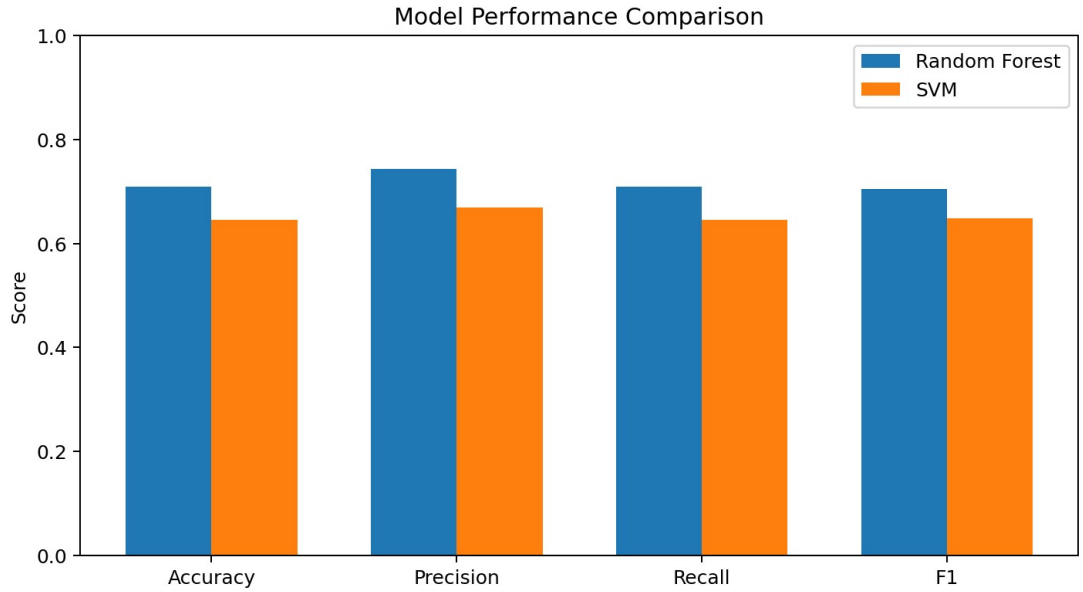

Figure 2. Random Forest scored higher on all four metrics.

Interpretation and Deployment

Random Forest was the better model with accuracy 0.710. Sunflower was the easiest class because it had a strong visual pattern. A confusion matrix showed more mix-ups for pizza and dalmatian. The model can be deployed by saving it with joblib and serving predictions through Flask or FastAPI with the same preprocessing pipeline.

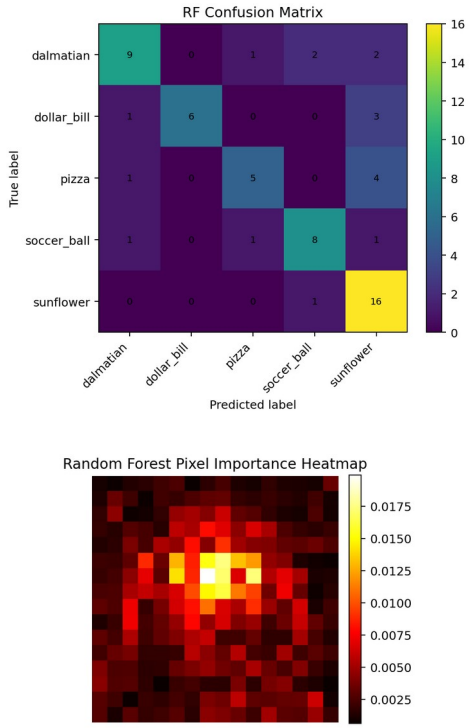

Figure 3. Random Forest confusion matrix and pixel-importance heatmap.

Prediction Example and Conclusion

The new-image prediction function applies grayscale conversion, resizing, normalization, and flattening before classification. In the test run, the example image was correctly predicted as 'dalmatian'. This project demonstrates a full image-classification workflow in Google Colab, including preprocessing, hyperparameter tuning, evaluation, interpretation, and deployment planning.